

Global Superstore SQL Data Analysis

Sales & Business Performance Analysis · MySQL · 2011–2014

48,361

Records Analyzed

\$7.81M

Total Sales

\$711K

Total Profit

4 Yrs

Analysis Period

Dataset

Global Superstore — a widely used benchmark retail dataset covering orders, customers, products, and geographies across multiple regions and categories.

Prepared by

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Tool: **MySQL** · Dataset: Global Superstore · Period: January 2011 – December 2014

Project Overview

Objective

Analyze Global Superstore sales data using SQL to understand business performance and identify important trends and opportunities across categories, regions, and time periods.

Key Areas Analyzed

→ Data Quality & Integrity

Validating completeness and consistency of the dataset before analysis.

→ Sales & Profit Performance

Category-level and product-level revenue and margin analysis.

→ Customer & Regional Analysis

Geographic performance and customer segmentation insights.

→ Discount & Shipping Impact

Relationship between discount levels, shipping modes, and profitability.

→ Yearly Growth Trends

Year-over-year sales and profit trajectory from 2011 to 2014.

SQL Skills Demonstrated

Core Queries

SELECT, WHERE, GROUP BY, ORDER BY, HAVING

Aggregations

SUM, AVG, COUNT, MAX

Advanced SQL

DISTINCT, JOIN, CTE, Window Functions, LAG

📌 Window functions and LAG were used to compute year-over-year growth rates directly in SQL — a key analytical technique for time-series business data.

Data Overview & Quality

795

Unique Customers

10,004

Unique Products

\$7.81M

Total Sales

\$711K

Total Profit

Dataset Summary

Attribute	Value
Total Records	48,361
Unique Customers	795
Unique Products	10,004
Date Range	Jan 1, 2011 – Dec 31, 2014
Total Sales	\$7,811,456
Total Profit	\$711,201.73

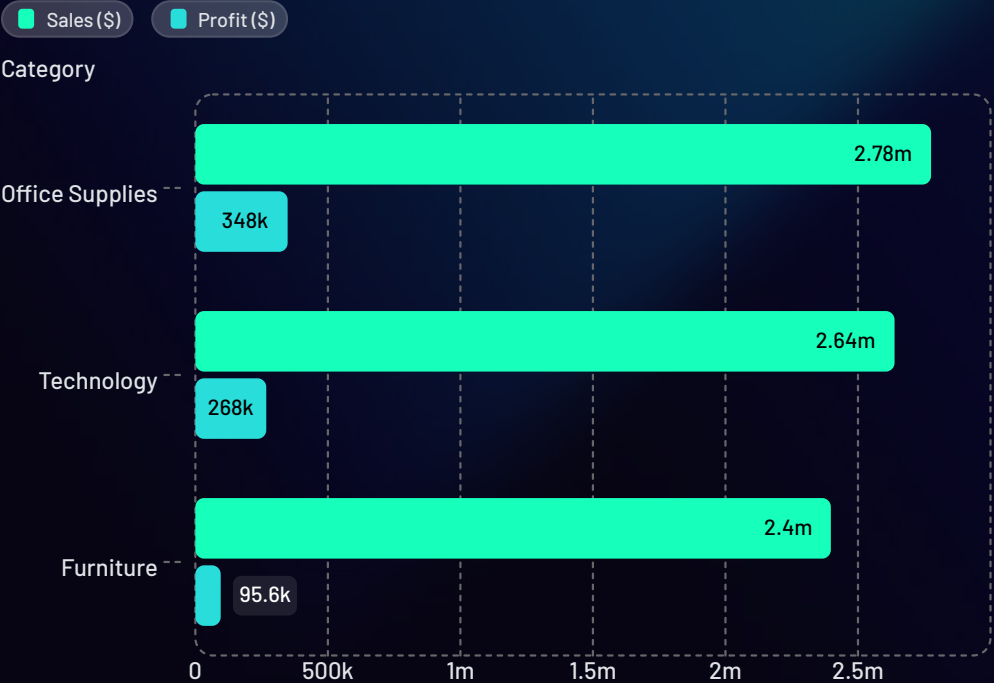
Data Quality Findings

Check	Result
NULL values (key columns)	0
Negative sales records	0
Invalid quantity records	0
Loss-making records	12,045
Distinct discount values	27 (range: 0–0.85)

📌 The dataset passed all integrity checks. The 12,045 loss-making records are treated as business-performance findings, not data errors – they represent a meaningful analytical signal.

Sales & Product Performance

Category Performance



Office Supplies leads in both sales and profit. Furniture generates strong revenue but delivers the lowest profit margin of the three categories — a key area for pricing review.

Top Products by Sales

- 1 Eldon File Cart, Single Width**
\$31,319 in sales
- 2 Rogers File Cart, Single Width**
\$22,645 in sales
- 3 Tenex File Cart, Single Width**
\$20,778 in sales

Top Products by Profit

- 1 Fellowes Lockers, Wire Frame**
\$4,342.02 profit
- 2 Rogers Locker, Single Width**
\$4,031.45 profit
- 3 Eldon Lockers, Industrial**
\$3,950.52 profit

Yearly Performance & Growth Trends



Year-over-Year Sales Growth

2012 +21.79%	2013 +25.35%
2014 +26.67%	

Growth Narrative

Sales and profit increased consistently across all four years. The growth rate itself accelerated year-on-year – from +21.79% in 2012 to +26.67% in 2014 – indicating compounding business momentum. From 2011 to 2014, total sales grew by **93%**, with 2014 delivering the strongest performance in both absolute revenue and growth rate.

Regional, Discount & Shipping Analysis

Top Regions by Sales & Profit

Region	Sales	Profit
Central	\$1,803,196	\$169,187
South	\$1,028,363	\$63,041
North	\$790,546	\$106,622

Most Profitable Locations

- 1

England

\$49,869.90 profit
- 2

California

\$46,630.31 profit
- 3

New York

\$32,586.97 profit

Discount Impact on Profitability

Low Discounts

Generally associated with positive profit margins across categories.

High Discounts (0.80–0.85)

Correlated with losses — the highest discount tier consistently produced negative margins.

Shipping Mode Performance

Mode	Sales	Profit
Standard Class	\$4,704,730	\$433,428
Second Class	\$1,577,232	\$141,806
First Class	\$1,131,715	\$90,489
Same Day	\$397,779	\$45,477

Standard Class dominates both volume and profitability, accounting for over 60% of total sales.

Key Business Insights & Conclusion

Overall Performance

\$7.81M in sales and \$711.2K in profit across 48,361 records demonstrate solid, scalable business performance over four years.

Category Profitability Gap

Office Supplies leads in profit (\$348K); Furniture shows a significant margin gap despite \$2.4M in sales — a clear signal for pricing and cost review.

Consistent Growth Trajectory

Sales grew 93% from 2011 to 2014, with the YoY growth rate accelerating each year — peaking at +26.67% in 2014.

Regional Concentration

Central region leads in sales; England and California lead in profitability. North delivers strong margins relative to its sales volume.

Discount Strategy Risk

High discounts (0.80–0.85) consistently produced losses. Discount discipline is directly linked to margin health across all categories.

Loss-Making Records

12,045 loss-making transactions warrant targeted review for pricing, discount policy, and cost optimization opportunities.

Conclusion

This SQL analysis demonstrates how structured querying reveals critical business patterns: category profitability gaps, regional performance variation, and the direct relationship between discount strategy and margins. These insights support strategic decisions on product mix, regional focus, and pricing optimization — translating raw data into actionable business intelligence.

Analytical Approach

01

Data Validation

Integrity checks across all key columns before analysis.

02

Exploratory Analysis

Aggregations by category, region, year, and product.

03

Advanced SQL

CTEs and window functions for growth rate computation.

04

Business Insight

Translating findings into strategic recommendations.